

Operators in Java 1, 2, 3 and 4:

Assignment, Sign, Arithmetic and Arithmetic Assignment

Lesson 1

Operators At A Glance

1. Assignment operator: = : To assign (store) a value to a variable
2. Sign operators: + - : To indicate positive and negative signs
3. Arithmetical operators: + - * / % : For arithmetical calculations
4. Arithmetic Assignment operators: += -= *= /= %= : Short hand for arithmetic and assignment
5. Increment and Decrement operators: ++ -- : To increment by 1 and decrement by 1
6. Concatenate operator: + : To join a string, number, character etc. to a string
7. Relational operators: > >= < <= == != : To check a condition
8. Logical operators: ! && || : For multiple conditions checkings
9. Conditional operators: ? : : To select an expression according to a condition

Forms of Operators : Unary, Binary and Ternary

Operator Precedence : Precedence and Associativity

Operator, Operands and Expression

1. Define **operator** with an example.

Operators are symbols that do some kind of **operations** (actions) on one or more **operands**.
E.g.: a = 5. Here = is an operator. The **a** and **5** are operands. The = assigns (stores) value 5 to the variable **a**. The = is the assignment operator.

2. Define **operands** with an example.

Operands are variables or literals (values) on which a specified operation is carried out.
E.g.: **a** and **5** are operands in **a=5**. The = is an operator. The = operator **operates on a** and **5**, i.e., the value **5** assigns to **a**.

3. Define an **expression** with an example and describe.

Expression is a **valid combination** of operands (variable or literal) and operator.

E.g.1: a+b Here a and b are operands. The + is the operator.

E.g.2: a+5 Here a and 5 are operands. The + is the operator.

Note: The **a+b=s** is **not an expression** because it is not valid. The valid is **s=a+b**

Answer the following:

State which are the following: **Operator** or **Operand** or **Expression**

- (1) m%2 (2) m (3) 2 (4) % (5) a/b (6) +

Answer

- (1) Expression (2) Operand (3) Operand (4) Operator (5) Expression (6) Operator

1. Assignment Operator =

1. The = is assignment operator; it is used to **assign (store)** a value to a variable. E.g.: a=5;

Right side of the = is executed first. E.g.: **a=3+4**; Here 3+4 is executed first. It is 7. Then 7 is stored to **a**. So the value of **a** is **7**. The **3+4=a** is an **invalid expression**.

2. Write output:

```
a) int x,y;
   x=y=7;
   System.out.println("x="+x);
   System.out.println("y="+y);
```

Answer

x=7

y=7

b) `int a=5, b=3;`

`a=b;`

`b=a;`

`System.out.println("a="+a);`

`System.out.println("b="+b);`

Answer

`a=3`

`b=3`

c) `int a=5, b=3;`

`int t=a;`

`a=b;`

`b=t;`

`System.out.println("a="+a+"\nb="+b);`

Answer

`a=3`

`b=5`

3. Correct the errors if any. (Write corrected statement only)

a) `int a=5,b=3,c;`

`a-b=c;`

answer

`c=a-b;`

2. Sign Operators + - (Unary + and Unary -)

1. Define sign operators with examples.

The **+** and **-** with a single variable or value are sign operators. E.g.: The **+a** means positive **a**. The **-a** means negative **a**. The **+5** means positive **5**. The **-5** means negative **5**.

2. Write output

`int x=7,y=-2,z=5,a=0;`

`System.out.println(-x);`

`System.out.println(-y);`

`System.out.println(-(-z));`

`System.out.println(-a);`

Answer

`-7`

`2`

`5`

`0`

3. Write values of c and d:

`int a=5,b=3,c,d;`

`c = a + -b;`

`d = a - -b;`

Answer:

`c=2`

`d=8`

3. Arithmetical Operators: + - * / %

1. The arithmetic operators are used for arithmetic calculations:

+	addition	To find sum
-	subtraction	To find difference
*	multiplication	To find product
/	division	To find quotient
%	modulus	To find the remainder

2. Write output.

a) `int a=7,b=2;`

`System.out.println(a+b);`

`System.out.println(a-b);`

`System.out.println(a*b);`

`System.out.println(a/b);`

`System.out.println(a%b);`

Output:

9

5

14

3 //Integer quotient because both values are integers

1 //Integer remainder

b) int a=2,b=4;

System.out.println(a-b);

System.out.println(a/b);

System.out.println(a%b);

Output

-2

0 //No fractional part. Small integer is divided by larger integer: Output is 0

2 //Small integer is mod by larger integer: Output is small integer

c) double a=5,b=2;

System.out.println(a);

System.out.println(b);

Output:

5.0 //Note the data type **double**. There is fractional part .0

2.0 //double type output

d) double a=7,b=2;

System.out.println(a/b);

System.out.println(a%b);

Output:

3.5 //because 7.0/2.0

1.0 //There is remainder

f) int a=7,b=2;

double c=a/b;

System.out.println(c);

Output:

3.0 //because a and b are integers. 7/2 is 3
when it stores to double it becomes 3.0

e) double a=2,b=4;

System.out.println(a/b);

System.out.println(a%b);

Output:

0.5

2.0

g) int a=2,b=4;

double c=a/b;

System.out.println(c);

Output

0.0 //because a and b are integers. 2/4 is 0
when it stores to double it becomes 0.0

Operator Precedence and Associativity

1. Define **operator precedence** with an example.

Operator precedence refers to the **order of evaluation** among operators in an **expression with different operators**. E.g.: Among operators = + /, the / has higher precedence, next + and last is =.

2. Write the value of a. $a = 6 + 8 / 2$.

$a = 6 + 8 / 2$

$a = 6 + 4$

$a = 10$

3. Define **operator associativity** with an example.

Associativity refers to the **order of evaluation** among **operators having same precedence**. The evaluation can be either from **left to right** or **right to left**. Operators * / % have same precedence. **Associativity** is left to right i.e., first operator executes first second one secondly ... last one lastly. The + - also have same precedence, associativity is left to right.

Example of left to right:

Write value of a. $a = 6 \% 8 * 2 / 3 * 5$.

$a = 6 \% 8 * 2 / 3 * 5$

$a = 6 * 2$

$a = 12 / 3$

$a = 4 * 5$

$a = 20$

4. Write value of a. $a = 6 + 8 - 4 + 5$.

$$a = 6 + 8 - 4 + 5$$

$$a = 14 - 4 + 5$$

$$a = 10 + 5$$

$$\underline{a = 15}$$

Example of right to left:

5. Write value of a and b of the expression: $a = b = 5$

$a = b = 5$ At first $b = 5$. Then $a = b$ i.e., $a = 5$, So,

Value of $a = 5, b = 5$

6. Write value of a and b:

int a=5, b=3;

a=b=a+b;

Answer

Working

$$a=b=a+b$$

$$a=b=5+3$$

$$a=b=8$$

$$a=8$$

Output:

a=8, b=8

The () Gives Higher Precedence

7. Write use of () in an expression.

The () in an arithmetic expression is used to give higher precedence to an operator that has lower precedence.

$5*3+4$ is 19. The $5*3$ evaluated first, it is 15, then $15+4$. It is **19**.

$5*(3+4)$ is 35. The $(3+4)$ is evaluated first, it is 7, then $5*7$. It is **35**.

8. Write output.

1) $2+4+6/3$

2) $(2+4+6)/3$

Answer:

1) 8

2) 4

9. Write value of the given expression: $a = 80 - 6 * 9 \% 8 / (3 + 1) \% 4$. Write all steps.

$$80 - \underline{6 * 9} \% 8 / (3 + 1) \% 4$$

At first $6*9$

$$80 - \underline{54 \% 8} / (3 + 1) \% 4$$

It is 54. Then $54\%8$

$$80 - 6 / \underline{(3 + 1)} \% 4$$

It is 6. Then $3+1$

$$80 - \underline{6 / 4} \% 4$$

It is 4. Then $6/4$

$$80 - \underline{1 \% 4}$$

It is 1. Then $1\%4$

$$\underline{80 - 1}$$

It is 1. Then $80-1$

$$\underline{79}$$

It is 79

$$\underline{a=79}$$

10. Write value of the given expression: $a = 50 + 6 \% 9 / 2 * (3 + 1) \% 4$. Write all steps.

$$50 + \underline{6 \% 9} / 2 * (3 + 1) \% 4$$

At first $6\%9$

$$50 + \underline{6 / 2} * (3 + 1) \% 4$$

It is 6. Then $6/2$

$$50 + 3 * \underline{(3 + 1)} \% 4$$

It is 3. Then $3+1$

$$50 + \underline{3 * 4} \% 4$$

It is 4. Then $3*4$

$$50 + \underline{12 \% 4}$$

It is 12 Then $12\%4$

$$\underline{50 + 0}$$

It is 0. Then $50+0$

$$\underline{50}$$

It is 50

$$\underline{a=50}$$

Division and Modulus (/ %) on Zero

1. `int a=0,b=2;`
`int x=a/b;`
`System.out.println(x);`
Answer:
0
2. `int a=0,b=2;`
`int x=a%b;`
`System.out.println(x);`
Answer:
0
3. `int a=5,b=0;`
`int x=a/b;`
`System.out.println(x);`
Answer:
 Division by zero error (Runtime error)
4. `int a=5,b=0;`
`int x=a%b;`
`System.out.println(x);`
Answer:
 Division by zero error (Runtime error)
5. `double a=0,b=2;`
`double x=a/b;`
`System.out.println(x);`
Answer:
0.0
6. `double a=0,b=2;`
`double x=a%b;`
`System.out.println(x);`
Answer:
0.0
7. `double a=5,b=0; double x=a/b;`
`System.out.println(x);`
Answer:
Infinity
8. `double a=5,b=0; double x=a%b;`
`System.out.println(x);`
Answer:
NaN
11. Write value of x: Write all steps.
`int a=78,b=5,c=4,d=7,e=6,f=2,g=3;`
`int x = a - b * c % d / (e + f) % g;`
 $x = 78 - 5 * 4 \% 7 / (6+2) \% 3$
 $x = 78 - 20 \% 7 / (6+2) \% 3$
 $x = 78 - 6 / (6+2) \% 3$
 $x = 78 - 6 / 8 \% 3$
 $x = 78 - 0 \% 3$
 $x = 78 - 0$
x=78
12. Write value of x: Write all steps.
`int a=48,b=5,c=8,d=3,e=2,f=6,g=7;`
`int x = a + b % c / d * (e + f) % g;`
 $x = 48 + 5 \% 8 / 3 * (2+6) \% 7$
 $x = 48 + 5 / 3 * (2+6) \% 7$
 $x = 48 + 1 * (2+6) \% 7$
 $x = 48 + 1 * 8 \% 7$
 $x = 48 + 8 \% 7$
 $x = 48 + 1$
x=49

4. Shorthand Arithmetic Assignment Operators += -= *= /= %=

1. Define shorthand arithmetic assignment operators

The shorthand arithmetic operators are the combination of arithmetic and assignment operators. It is used **to modify the existing value in a variable**.

2. Examples of Arithmetic Assignment Operators

Initial value of **a = 5** for each example.

Operator	Example	Expansion	Working	Result
+=	a+=2	a=a+2	a=5+2	a=7
-=	a-=2	a=a-2	a=5-2	a=3
=	a=2	a=a*2	a=5*2	a=10
/=	a/=2	a=a/2	a=5/2	a=2
%=	a%=2	a=a%2	a=5%2	a=1

3. Write values of x: int x=7, y=3;

- x+=y;
- x-=y;
- x*=y;
- x/=y;
- x%=y;

Answers

- 10
- 4
- 21
- 2
- 1

4. The = and += have same precedence but associativity is right to left.

a) Write values of a and b:

```
int a=2,b=3;
b=a+=b;
```

Working
b=a+=b
b=a+a+b
b=a=2+3
b=a=5
b=5

a=5, b=5

b) Write values of m and n:

```
int m=4,n=5;
m=n+=m;
```

Working
m=n+=m
m=n=n+m
m=n=5+4
m=n=9
m=9

m=9, n=9

c) Write values of a and b:

```
int a=2,b=3;
b+=a=b;
```

Working
b+=a=b
b=b+a=b
b=b+a=3
b=3+3
b=6

a=3, b=6

d) Write values of **m** and **n**:

```
int m=4,n=5;
m+=n=m;
```

Working

```
m+=n=m
m=m+n=m
m=m+n=4
m=4+4
m=8
```

m=8, n=4

5. The += executes right to left

a) Write values of **m** and **n**.

```
int m=2,n=3;
n+=m+=n;
```

```
n+=m+=n
n=n+m=m+n
  m=2+3
  m=5
n=n+5
n=3+5
n=8
```

m=5, n=8

b) Write values of **x** and **y**.

```
int x=4,y=5;
y+=x+=y;
```

```
y+=x+=y
y=y+x=x+y
  x=4+5
  x=9
y=y+9
y=5+9
y=14
```

x=9, y=14

6. Arithmetic operators executes first then arithmetic shorthand operator

1) `int a=2;`
`a*=a+5;`

```
a*=a+5
a = a * (a + 5)   Practice using ( )
  = 2 * (2 + 5)
  = 2 * 7
  = 14
```

a=14

2) `int a=5;`
`a/=a-2;`

```
a/=a-2
a = a / (a - 2)
  = 5 / (5 - 2)
  = 5 / 3
  = 1
```

a=1

3) `int a=5;`
`a%=a-2;`

```
a%=a-2
a = a % (a - 2)
  = 5 % (5 - 2)
  = 5 % 3
  = 2
```

a=2

4) `int a=5;
a+=a-2;`

$$\begin{aligned} a &+= a - 2 \\ a &= a + (a - 2) \\ &= 5 + (5 - 2) \\ &= 5 + 3 \\ &= 8 \end{aligned}$$

a=8
5) `int a=5;
a-=a+2;`

$$\begin{aligned} a &-= a + 2 \\ a &= a - (a + 2) \\ &= 5 - (5 + 2) \\ &= 5 - 7 \\ &= -2 \end{aligned}$$

a=-2
6) `int a=5,b=2;
a*=a-b;`

$$\begin{aligned} a &*= a - b \\ a &= a * (a - b) \\ &= 5 * (5 - 2) \\ &= 5 * 3 \\ &= 15 \end{aligned}$$

a=15
7) `int a=5,b=2;
a/=a-b;`

$$\begin{aligned} a &/= a - b \\ a &= a / (a - b) \\ &= 5 / (5 - 2) \\ &= 5 / 3 \\ &= 1 \end{aligned}$$

a=1
8) `int a=5,b=2;
a%=a-2;`

$$\begin{aligned} a &\%= a - b \\ a &= a \% (a - b) \\ &= 5 \% (5 - 2) \\ &= 5 \% 3 \\ &= 2 \end{aligned}$$

a=2

Difference of Numbered and Non-Numbered Problems

7. Write output: (Questions are numbered. So each are separate questions.)

`int x=5,y=3;` (For all questions)

1. `System.out.println(x-=y);`
2. `System.out.println(y+=x);`
3. `System.out.println(x=y/=x);`
4. `System.out.println(y*=x);`
5. `System.out.println(x%=y);`

Output:

- 1) 2
- 2) 8
- 3) 0
- 4) 15
- 5) 2

Working

$$\begin{aligned} 2 &= 5 - 3 \\ 8 &= 3 + 5 \\ 0 &= 0 = 3 / 5 \\ 15 &= 3 * 5 \\ 2 &= 5 \% 3 \end{aligned}$$

8. Write output: (Questions are not numbered. So it is a single question.)

`int x=5,y=3;
System.out.println(x-=y);
System.out.println(y+=x);
System.out.println(x=y/=x);
System.out.println(y*=x);
System.out.println(x%=y);`

Output:

- 2
- 5
- 2
- 4
- 2

Working

$$\begin{aligned} 2 &= 5 - 3 \\ 5 &= 3 + 2 \\ 2 &= 2 = 5 / 2 \\ 4 &= 2 * 2 \\ 2 &= 2 \% 4 \end{aligned}$$